INCH-POUND

MIL-E-85082/6

16 November 1992

MILITARY SPECIFICATION SHEET

ENCODER, SHAFT POSITION TO DIGITAL
CONTACT, ABSOLUTE
TYPE 11KCJ10S05A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Encoders, Shaft Angle to Digital, described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-E-85082.

TABLE I. Requirements.

Requirements	Value	Unit	Tolerance
Code and scan	Natural Binary, V-scan		
Capacity	2^{10}	counts	maximum
Resolution	2^{5}	counts/turn	
Full count revolutions	1	turn	
Voltage interrogation	5.0	volts DC	maximum
Contact current	5.0	mA/bit	maximum
Starting torque	0.4	oz-in	minimum
Running torque	0.4	oz-in	maximum
Operating speed	200	rpm	maximum
Slew speed	1000	rpm	maximum
Radial play	0.0006	inch	maximum
Shaft runout	0.0008	inch	maximum
End play	0.0020	inch	maximum
Perpendicularity	0.0020	inches	maximum
Operating temperature	-65° to +85°	degrees C	min-max
Transition accuracy	±1/2	bit	
Moment of inertia	5.0	$gm\text{-}cm^2$	maximum
Weight (without leads)	2.0	oz	maximum
Life	10,000,000	revolutions	

AMSC - N/A FSC 5990

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TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±0.015	E +0.0000 -0.0005	G Maximum	Shaft Style	Terminal End	U +0.0000 -0.0003
M85082/6-01A	.500	.6250	1.000	Plain	Wire Leads	.1875

1/ Part identifying number M85082/6-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Wire termination identification.

FUNCTION	COLOR
OUTPUT A	BLUE
OUTPUT B	BLUE/WHITE
OUTPUT C	RED
OUTPUT D	RED/WHITE
COMMON	WHITE

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NOTES:

1. Wire leads shall be in accordance with MIL-W-16878, type E-26.
2. Dimension C shall be 1.060 with a tolerance of ±0.003.
3. Dimension D shall be 1.000 with a tolerance of +0.000/-0.001.
4. Dimension H shall be 0.062 with a tolerance of +0.000/-0.003.
5. Dimension I shall be 0.062 with a tolerance of +0.000/-0.005
6. Dimension J shall be 0.095 with a tolerance of +0.000/-0.005.
7. Dimension K shall be 0.055 with a maximum tolerance.
8. Dimension L shall be 0.975 with a tolerance of +0.000/-0.005.
9. Rotation for increasing count shall be clockwise.
10. Lead exit shall be sealed.
11. Radial play shall be measured with an 8 oz reversing load applied no more than 0.25 inch from bearing face.
12. End play shall be measured as total movement with a 1 lb reversing load.

Custodians:
Navy - AS
Army - AR
Air Force - 85

Review activities:
DLA - ES

Preparing activity:
Navy - AS

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